The LM386: An Exercise in Reasonable Expectations

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Abstract

This report presents the design, implementation, and evaluation of a stereo audio amplifier using the LM386 integrated circuit. The objective was to produce a circuit capable of amplifying a low-level audio signal to a level suitable for driving headphones, while meeting constraints on gain, input impedance, and component availability. A two-stage architecture was designed. Due to practical limitations, the intended input buffer stage was not successfully implemented.

Experimental results showed that the amplifier produced a clear and sufficiently loud output with minimal audible distortion. The initial gain was measured below the required value of 20, but was corrected through adjustment of the LM386 gain configuration, achieving a final gain of approximately 20.27. However, the absence of the buffer stage resulted in a significantly reduced input impedance, meaning the full design specification was not met.

Keywords: *LM741, LM386, audio, amplifier, gain, headphones, emitter follower*

1 Introduction

1.1 Fundamentals of signal amplification

Signal amplification is the process of increasing the magnitude of a signal by using energy drawn from an external supply, while ideally preserving the information carried by the original waveform (Cheung 2019). In theory, this means producing an output that is a scaled version of the input where the degree to which the input is scaled is characterised by a quantity called gain. In practice, real amplifiers introduce non-ideal effects such as noise and distortion (Niknejad and Muller 2016).

1.2 Practical applications

In communication systems, amplifiers are used to boost signals for transmission over long distances and to recover attenuated signals on the receiving end. In sensing applications, amplification allows small signals, such as those from microphones or sensors, to be analysed and measured without specialised hardware. In audio systems, amplification is required to increase input power to levels capable of driving headphones or speakers.

1.3 Requirements in an audio context

Amplification must increase the signal to a sufficient level, while preserving the original waveform. This requires low distortion, low noise, and adequate bandwidth across the audible range of 20 Hz to 20 kHz (Kinsler et al. 2000; Luecke 2005). Input and output impedance are also particularly important in audio applications, since the amplifier must avoid loading the source while still being able to drive a low-impedance load.

1.4 Report structure

This report follows the design, implementation and evaluation of a basic audio amplifier. Objectives and constraints are discussed in the next section, followed by a description of the schematic design and component selection. The implementation section details the construction and practical considerations, after which the experimental methodology and results are presented. Finally, the results are discussed, with conclusions drawn regarding the performance and limitations of the design.

2 Aim, Objectives & Constraints

The objective of this experiment was to design and implement a functional audio amplifier capable of increasing a low-level input signal to a level suitable for driving headphones, while maintaining acceptable signal quality and satisfying the specification. The objectives were:

- Design an appropriate topology using at least one LM741 or LM386 IC.
- Achieve a minimum gain of 20.
- Achieve a minimum input impedance of 100 kΩ.
- Drive an output load of 32 Ω.
- Construct and test the amplifier using components available in the laboratory.

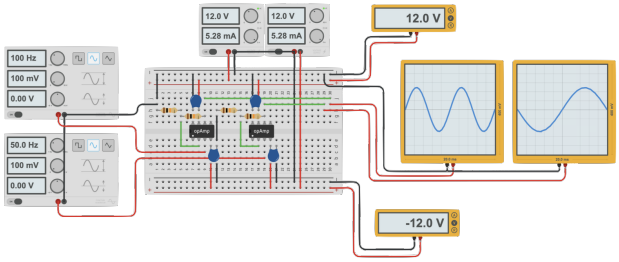


Figure 1: *Dual channel LM741 breadboard simulation*

The design was subject to several practical constraints. The final design required to operate from a supply voltage of $\pm 12\,\text{V}$. Capacitor selection was restricted: initially the only available unpolarised capacitors were 4.7 µF and 47 µF, and polarised capacitors were forbidden due to their tendency for spontaneous disassembly in amateur circuits[1].

It should be noted that the restriction on available capacitor values was not known during the design stage discussed below. As a result, the circuit was further modified during build according to component availability.

3 Design

3.1 Research and component selection

It was understood from the outset that neither the LM741 nor the LM386 would be suitable as a standalone solution: the LM741 cannot drive low-impedance loads, while the LM386 does not meet the input impedance requirement (Texas Instruments 2023; Texas Instruments 2015). The LM741 was selected for initial experimentation due to its availability for simulation in Tinkercad.

A single-channel non-inverting amplifier was designed based on the typical application circuit in the LM741 datasheet, with a target gain of approximately 21. This design was simulated in Tinkercad using a signal generator and oscilloscope to verify expected gain behaviour. Following successful validation, the design was extended to a dual-channel configuration to support stereo input via a TRS jack, with both channels implemented identically and illustrated in Figure 1.

Further analysis confirmed the expected limitations of each component. The LM741 is not designed to drive low-impedance loads (recommended $\geq 2\,\text{k}\Omega$), making it unsuitable for directly driving 32 Ω headphones (Lisleapex 2020). It also has a limited slew rate of approximately 0.5 V/µs (Texas Instruments 2015). Assuming a 10V peak signal, applying the relationship in Equation 1 gives a maximum undistorted frequency of 8 kHz, indicating high-frequency audio components would be subject to slew-induced distortion if used in the amplification stage (Analog Devices 2005).

$$\text{SR} \geq 2\pi f V_p \qquad \text{f} \leq \frac{0.5 \div 10^{-6}}{2\pi \times 10} \approx 8\ \text{kHz} \qquad (1)$$

Conversely, while the LM386 is well suited to driving low-impedance loads, its typical input impedance of 50 kΩ does not meet the specification (Texas Instruments 2023). However, it is a convenient choice for the amplification stage, as it provides a default gain of 20 (which can be increased to 200 with an appropriate external capacitor[2]).

As a result, a two-stage topology was adopted, combining a high-input-impedance stage with a dedicated power amplification stage, allowing each component to be used in a role better aligned with its strengths.

3.2 Final design

The final design adopts a two-stage stereo amplifier topology, with identical circuitry for the left and right channels. The signal path, shown in Figure 2, is as follows:

- Input signal enters via a TRS jack and is AC-coupled using a series capacitor, providing DC isolation
- An RC low-pass filter attenuates high-frequency interference
- The signal is buffered using an emitter follower stage, providing high input impedance
- A potentiometer is used to adjust signal amplitude prior to amplification
- The LM386 amplifies the signal, initially configured with a gain of 20
- A Zobel network is included at the output to stabilise the amplifier when driving the inductive headphone load
- A large output capacitor provides DC isolation, preventing the amplifier bias voltage from reaching the headphones

Due to limited breadboard space and wiring complexity, the input buffer stage was simplified from the originally proposed LM741 unity-gain configuration to a discrete emitter follower using a single NPN transistor. The 2SC1845 was selected as it is commonly used in audio applications and was available on-hand (NEC 1979). This preserves the required high input impedance while improving build practicality.

[1] Reverse bias is not well tolerated and the failure mode is typically immediate and energetic.

[2] The musical consequences of doing so are left to the experimenter.

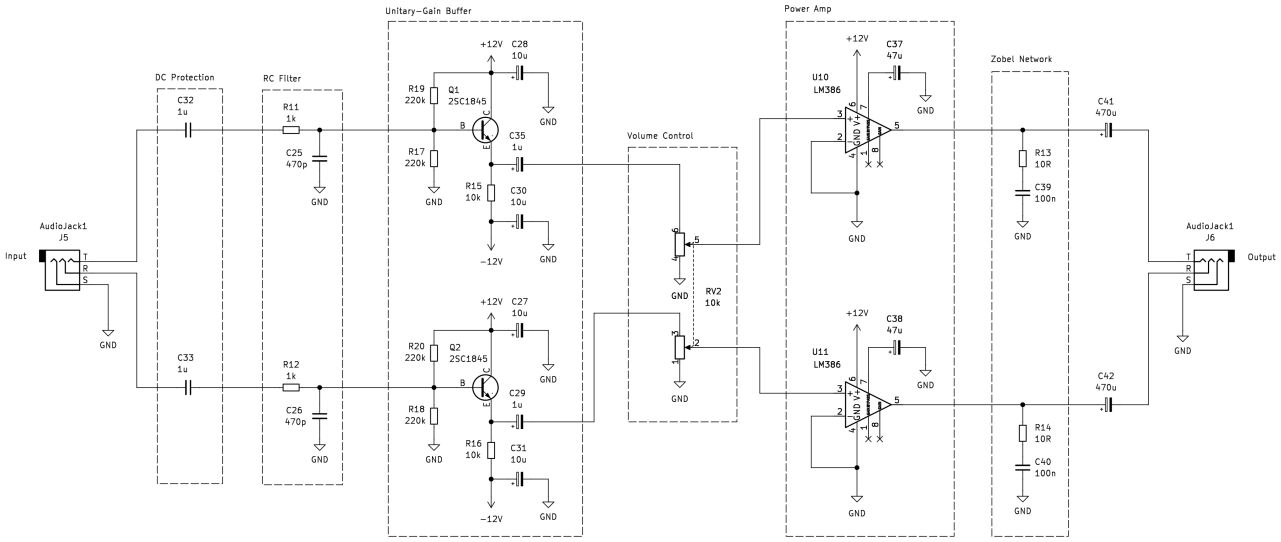


Figure 2: *Two stage stereo amplifier design*

The cutoff frequency of the input RC low-pass filter was selected according to Equation 2 with $R = 1\,\text{k}\Omega$ and $C = 470\,\text{pF}$:

$$f_c = \frac{1}{2\pi RC} \approx 339\,\text{kHz} \tag{2}$$

This is well above the audible range, ensuring minimal attenuation of the desired signal.

The Zobel network was included at the output to improve stability when driving the inductive headphone load (Self 2009). A capacitor value of 50 nF was selected in accordance with the LM386 datasheet recommendation, with a series resistor of $10\,\Omega$ selected according to Equation 3:

$$f_z = \frac{1}{2\pi RC} \approx 318\,\text{kHz} \tag{3}$$

Again, this frequency is far above the audible range, so the network has negligible effect on the desired audio signal.

4 Implementation

The circuit was initially constructed in a simplified form to confirm amplification was, at minimum, occurring. A single-channel configuration was implemented consisting of the RC input filter, potentiometer, LM386 amplification stage, and Zobel network. DC protection, buffering, and output coupling capacitors were omitted, as they were not required to verify initial amplifier operation.

Following construction, the circuit was tested using a signal generator and oscilloscope to verify output behaviour. The amplifier produced a clear amplified signal, but the measured gain was lower than the required value of 20. This was not unexpected, as the default LM386 gain of 20 is applied after attenuation of the input signal by the RC filter and potentiometer. The measured gain was within reasonable bounds (approximately 18), so development proceeded to the dual-channel build with the expectation that the gain would later be increased. This is discussed further in the testing section.

The design was then extended to a dual-channel configuration to support stereo input, pictured in Figure 3. Due to limited component availability, two single-gang potentiometers were used in place of a dual-gang potentiometer, allowing independent control of each channel at the cost of some symmetry. This configuration implemented all components of the design, with the exception of the emitter follower stage, whose eventual absence is discussed later.

During early testing, intermittent faults were observed, consistent with instability in the power supply. This was resolved by adding a 47 µF decoupling capacitor directly across the 12 V supply rails. Following this modification, circuit operation stabilised.

5 Testing & Evaluation

For all oscilloscope measurements presented in this section, the three channels are assigned as follows:

- Chan 1 (Blue): Input signal
- Chan 2 (Cyan): Output signal (left channel)
- Chan 3 (Magenta): Output signal (right channel)

5.1 Subjective listening

Initial testing was conducted subjectively by comparing the audio output with and without amplification. The output volume on the laptop was first set to

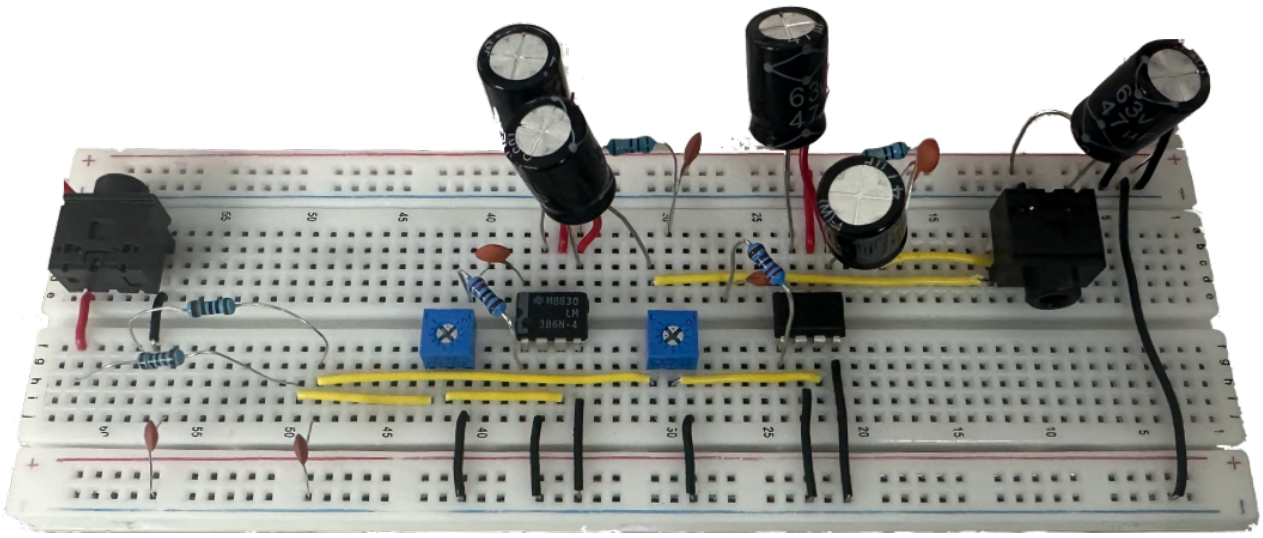


Figure 3: *Dual-channel single stage amplifier build*

100%, and the Geoff Castellucci cover of *Big Bad John* (Jimmy Dean, 1961) was used as the test track. This track is commonly used for speaker testing due to its wide vocal range, deep bass presence, and overall production quality, despite not being a particularly strong musical recommendation.

The laptop output was then reduced to 10%, and the track was first listened to without amplification as the control sample, followed by playback through the amplifier. During control playback, the potentiometers were adjusted to verify balance of the left and right channels.

Without amplification, the track was audible but quiet. With amplification, the output was significantly louder and suitable for normal listening levels without requiring an increase in the source volume. The audio remained clear, with no immediately noticeable distortion. However, adjustment of the potentiometers during playback occasionally introduced noise, suggesting sensitivity to contact variation or interference.

Compared to direct playback from the laptop as the control sample, the amplified output was perceptibly flatter, with reduced bass response. This is consistent with the limitations of the circuit and component selection. The subjective listening results are detailed in Table 1.

Table 1: *Subjective audio response*

	Volume	Bass	Mid	Treble
Control	10%	Medium	High	High
Test	∼50-70%	Low	High	High

5.2 Stress test

A stress test was conducted to determine the maximum input signal before clipping occurred. The potentiometers were first adjusted to provide maximum gain, after which the amplitude of the input signal from the signal generator was gradually increased. The output was monitored using the oscilloscope until visible clipping was observed as shown in Figure 4.

Clipping occurred at an input voltage of approximately 0.8 V (corresponding to 1.6 V_{pp}), which is higher than the theoretical value of 0.6 V based on a gain of 20 and a 12 V supply. This discrepancy is attributed to voltage losses introduced by the input network, specifically the resistor used in the low-pass filter and the potentiometer, which together attenuate the signal reaching the amplifier stage.

Beyond this point, the output waveform deviated from the expected sinusoidal shape and took on a more interpretive form, indicating the amplifier had exhausted its ability to remain linear.

5.3 Amplification gain

The gain of the amplifier was measured using a sinusoidal input signal. At an input level of 232 mV_{pp}, the output voltage on both left and right channels was measured at 4.24 V_{pp}, corresponding to a gain of approximately 18.28. This is lower than the theoretical gain of 20, due to attenuation introduced by the RC input filter and the potentiometer resistance.

The LM386 provides a default gain of 20, set by an internal 1.35 kΩ resistor (Texas Instruments 2023). When an external gain resistor is added between pins 1 and 8, it operates in parallel with this internal

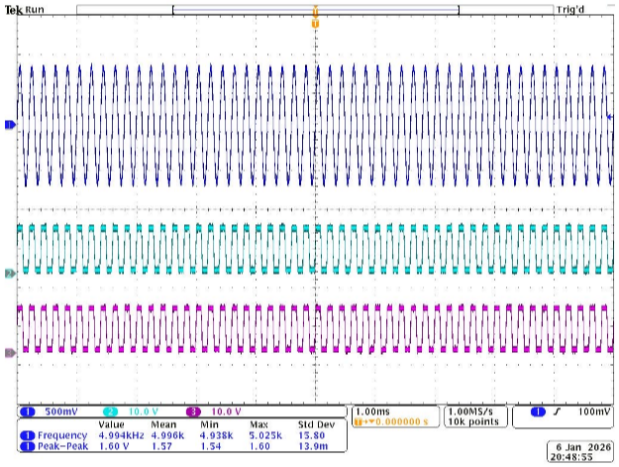


Figure 4: *Stress test indicating clipping of output signal*

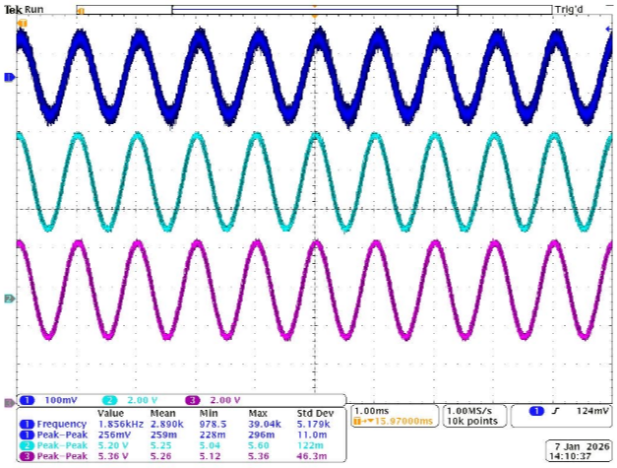


Figure 5: *20.27 output gain from a 30.4 amplification gain*

resistance, modifying the overall gain according to:

$$\text{Gain} = \frac{30000}{150 + (1350 \parallel R_{\text{gain}})} \tag{4}$$

where $\parallel$ denotes the parallel resistor combination. Selecting a resistor value of 2.2 kΩ gives:

$$\text{Gain} = \frac{30000}{150 + (1350 \parallel 2200)} \approx 30.4 \tag{5}$$

This value was intentionally chosen to exceed the required gain, compensating for losses elsewhere in the signal path. After implementing this modification on both channels, the measured gain increased to approximately 20.27, as shown in Figure 5.

This result demonstrates that while the nominal gain of the LM386 can be easily configured, the effective system gain is ultimately determined by the surrounding circuitry.

5.4 Input impedance

Due to time constraints, the emitter follower input buffer was not successfully implemented in the physical circuit. However, the configuration was simulated in Falstad and behaved as expected, as shown in Figure 6. This suggests that the design itself was sound, even if the breadboard implementation was less successful.

Without the buffer stage, the amplifier was expected to fail the input impedance requirement. The 10 kΩ volume potentiometer is effectively in parallel with the LM386 input impedance of approximately 50 kΩ, with the 1 kΩ resistor from the RC filter in series. The resulting best-case input impedance is given by Equation 6.

$$Z_{\text{in}} = (10\,\text{k}\Omega \parallel 50\,\text{k}\Omega) + 1\,\text{k}\Omega \approx 9.33\,\text{k}\Omega \tag{6}$$

If the emitter follower stage had been implemented successfully, the expected input impedance would have been substantially higher. The two base bias resistors form a voltage divider, whose equivalent resistance is given by Equation 7.

$$R_{\text{bias}} = R_1 \parallel R_2 = 110\,\text{k}\Omega \tag{7}$$

The emitter sees the parallel combination of the emitter resistor and the following stage load, approximated by Equation 8.

$$R_{\text{emit}} = 10\,\text{k}\Omega \parallel 10\,\text{k}\Omega = 5\,\text{k}\Omega \tag{8}$$

For a rank F 2SC1845, a typical current gain of $\beta \approx 500$ may be assumed (NEC 1979). The impedance seen looking into the base is therefore approximated by Equation 9.

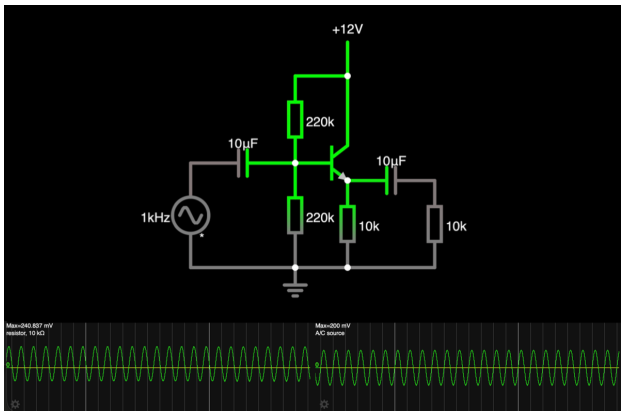


Figure 6: *Falstad simulation of emitter follower buffer*

$$Z_{\text{base}} \approx \beta R_{\text{emit}} = 500 \times 5\,\text{k}\Omega = 2.5\,\text{M}\Omega \tag{9}$$

This base impedance appears in parallel with the bias network, with the RC filter resistor again in series. The total buffered input impedance is therefore given by Equation 10.

$$Z_{\text{in}} = (R_{\text{bias}} \parallel Z_{\text{base}}) + 1\,\text{k}\Omega \approx 106.4\,\text{k}\Omega \tag{10}$$

According to Equation 10, the emitter follower stage should have allowed the design to meet the input impedance specification, assuming typical transistor behaviour and a more cooperative practical implementation.

5.5 Output power

The resistance of the headphones was measured as 32.4 Ω, and the output voltage across each channel was measured as 5.25 V_{pp}. Since AC power is calculated using the equivalent DC value that would produce the same heating effect in a resistive load, the RMS voltage must first be found (OpenStax 2022).

For a sinusoidal waveform, the RMS voltage is given by

$$V_{\text{rms}} = \frac{V_{pp}}{2\sqrt{2}} = 1.856\,\text{V} \tag{11}$$

The output power delivered to one headphone channel can then be calculated using Equation 12 where R_h is the headphone resistance.

$$P = \frac{(V_{\text{rms}})^2}{R_h} = 106.3\,\text{mW} \tag{12}$$

Therefore, the amplifier delivers approximately 106.3 mW per channel. As the circuit is dual-channel, the total output power is 212.6 mW.

6 Conclusions

The amplifier was successful in producing a clear and noticeably louder audio output, meeting its primary objective of basic audio amplification. Subjective listening tests showed that the output was suitable for normal listening levels, with minimal audible distortion and only limited hiss. However, the amplified output was perceptibly flatter than the direct laptop output, particularly in the bass response, indicating that signal quality was acceptable rather than exceptional.

The initial measured gain of the circuit was approximately 18.27, below the required value of 20. This was attributed to attenuation introduced by the RC filter and potentiometers. By adding 2.2 kΩ resistors to the LM386 gain pins, the measured gain was increased to approximately 20.27, satisfying the specification.

The main unresolved limitation was the input impedance. As the emitter follower buffer stage was not successfully implemented, the practical input impedance remained far below the required 100 kΩ. In the final build, the 10 kΩ potentiometer was effectively in parallel with the LM386 input impedance of 50 kΩ, with the 1 kΩ RC filter resistor in series, giving a best-case input impedance of approximately 9.33 kΩ. This means that, while the amplifier functioned well enough in practice, it did not satisfy the full design specification.

If the emitter follower stage were implemented correctly, the expected input impedance would increase to approximately 106 kΩ, allowing the design to meet the original requirement. Further work is therefore proposed, as is customary.

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